

Market Equilibrium as Purpose Fixed-Point: A Mathematical Theory of Coordination among Bounded Economic Agents

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Abstract

This paper develops a rigorous mathematical theory of market coordination among heterogeneous bounded agents. Taking the receiver calculus and agent triple (R, M, G) from the companion paper [32] as primitives, we extend to ensembles and prove ten principal results: (i) the *Belief Incompatibility Theorem* — agents with disjoint knowledge frameworks cannot share beliefs at any state; (ii) the *Common-Cell Convergence Theorem* — despite belief incompatibility, every agent in the ensemble can simultaneously attain the same action-cell; (iii) the *Multi-Agent Catalytic Composition Law* — the composite floor of n independent agents equals the product of individual floors divided by Σ^{n-1} ; (iv) the *Purpose Existence Theorem* — every independent ensemble possessing a cell of sufficient tolerance has a purpose, proved via the Banach contraction structure of the methodology without any convexity assumption; (v) *Purpose Stability* — the equilibrium cell is preserved under small perturbations of the ensemble; (vi) the ω -*Limit Theorem* — any agent flow starting in the reachability set converges to the purpose cell; (vii) the *Motivation Heterogeneity Theorem* — the composite floor depends only on per-agent floors, not on goal-content; (viii) the $\boxtimes$ *algebra* — a commutative monoid structure on market floors; (ix) *Market Information Efficiency* — precise necessary and sufficient conditions for the Efficient Market Hypothesis expressed as a Borel–Cantelli-type divergence condition; and (x) *Coordination without Common Knowledge* — replacing Aumann’s common-knowledge hierarchy with cell exteriority. Forty-five numerical experiments verify all claims to machine precision (maximum relative error $\leq 10^{-14}$).

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1 Introduction

1.1 The Coordination Problem

Markets coordinate agents whose beliefs are incompatible, whose goals oppose one another, and whose information is heterogeneous. Three classical frameworks address this coordination problem, each at the cost of assumptions that bounded agents cannot satisfy.

Nash equilibrium [29, 30]. A Nash equilibrium is a profile of strategies such that no agent can improve its payoff by unilateral deviation. Its existence proof — via Kakutani’s fixed-point theorem — requires compact convex strategy spaces, upper-semicontinuous best-response correspondences, and, in the epistemic interpretation, common knowledge of rationality. Bounded agents violate all three requirements: their strategy sets need not be convex, their best-response maps need not be continuous, and they cannot verify that others are rational or that this is common knowledge.

Arrow–Debreu competitive equilibrium [3, 9, 25]. A competitive equilibrium is a price vector p^* clearing all markets simultaneously. The Arrow–Debreu framework requires complete contingent claims markets (a market for every commodity in every state of the world), convex preferences and production sets, a Walrasian auctioneer who observes all excess demands, and the absence of externalities. None of these conditions holds in any real market.

Rational expectations equilibrium [27, 21]. A rational expectations equilibrium requires that agents form beliefs consistent with the true data-generating process. This amounts to demanding

that every agent knows the model — a requirement that, in the language of our companion paper, collapses each agent’s floor β_i to zero. The floor theorem of [32] proves this impossible for any bounded receiver.

The present paper derives coordination from a single primitive — the bounded receiver agent triple (R, M, G) — without imposing convexity, completeness, common knowledge, or model knowledge.

1.2 Main Contributions

- (i) **Belief Incompatibility Theorem** (theorem 3.2). Agents with representation-disjoint knowledge frameworks $K_i \cap K_j = \emptyset$ possess beliefs that are structurally incompatible at every state of the world. No isomorphism can bridge their decoded representations.
- (ii) **Common-Cell Convergence Theorem** (theorem 4.1). Despite full belief incompatibility, every agent in an ensemble can simultaneously attain the same action-cell $C \subseteq X$. Coordination occurs in outcome space, not in representation space.
- (iii) **Multi-Agent Catalytic Composition Law** (theorem 5.2). For an independent ensemble $\mathcal{E} = (A_1, \dots, A_n)$, the composite floor satisfies $S_b(A_1 \diamond \dots \diamond A_n) = \prod_{i=1}^n S_b(A_i) / \Sigma^{n-1}$.
- (iv) **Purpose Existence Theorem** (theorem 8.1). Every independent ensemble with composite floor $S_b(\mathcal{E}) < \Sigma$ and a cell of tolerance $\tau(C) > S_b(\mathcal{E})$ possesses a purpose C^* . The proof uses Banach contraction; no convexity is assumed.
- (v) **Purpose Stability** (theorem 8.5). The equilibrium cell C^* is preserved under ε -perturbations of the ensemble’s composite floor, provided $\varepsilon < \tau(C^*) - S_b(\mathcal{E})$.
- (vi) **ω -Limit Theorem** (theorem 9.2). Any continuous agent flow starting in $\text{Reach}(\mathcal{E}, C)$ converges to C^* : every ω -limit point lies in C^* .
- (vii) **Motivation Heterogeneity Theorem** (theorem 10.2). The composite floor depends solely on per-agent floors $S_b(A_i)$, not

on goal-contents (G_i, Act_i). Buyers and sellers with opposing goals compose to the same floor as agents sharing a goal.

- (viii) $\boxtimes$ **Algebra** (theorem 11.2). The triple $([0, \Sigma], \boxtimes, \Sigma)$ is a commutative monoid, where $f_1 \boxtimes f_2 := f_1 f_2 / \Sigma$. Catalytic composition is the monoid operation on floors.
- (ix) **Market Information Efficiency** (theorem 13.2). The Efficient Market Hypothesis holds (i.e. $S_b(\mathcal{E}) = 0$) if and only if $\sum_{i=1}^{\infty} \log(1/q_i) = \infty$, where $q_i = S_b(A_i)/\Sigma$. For any finite ensemble, efficiency is strictly partial.
- (x) **Coordination without Common Knowledge** (theorem 4.5). Ensemble coordination on cell C^* requires only that each agent's decoder-projection chain reaches C^* independently. No common knowledge of C^* , no iterated knowledge hierarchy, is needed — replacing the Aumann [4] framework with cell exteriority.

1.3 Organization

Section 2 introduces heterogeneous agent ensembles. Section 3 proves the Belief Incompatibility Theorem. Section 4 establishes Common-Cell Convergence and coordination without common knowledge. Section 5 develops the Multi-Agent Catalytic Composition Law. Section 6 analyzes reachability and market depth. Section 7 introduces the purpose functional and equilibrium definition. Section 8 proves existence, uniqueness, and stability. Section 9 proves the ω -Limit Theorem. Section 10 establishes motivation heterogeneity. Section 11 develops the $\boxtimes$ algebra. Section 12 studies the coalition lattice. Section 13 treats market information efficiency and the EMH. Section 14 compares with classical equilibrium theory. Section 15 presents the forty-five numerical experiments. Section 16 surveys connections to the literature. Section 17 states conclusions.

2 Heterogeneous Agent Ensembles

We recall from [32] the agent triple and its associated floor. An *agent* is a triple $A = (R, M, G)$ consisting of a *receiver* $R = (K, D, \Pi, \beta)$, a *methodology* M , and a *goal set* G . The receiver components

are: K the knowledge framework (representation alphabet), $D : X \rightarrow K$ the decoder, $\Pi : K \rightarrow 2^X$ the candidate-projection, and $\beta > 0$ the receiver floor. The agent's S -functional at state $x \in X$ targeting cell $C \subseteq X$ is $S(A, x; C) := \inf_{c \in \Pi(D(x))} d(c, C) + \beta$, and its floor is $S_b(A) = \beta \cdot S_b(M)/\Sigma$.

Definition 2.1 (Ensemble). *An ensemble is an ordered tuple $\mathcal{E} = (A_1, \dots, A_n)$ where each agent $A_i = (R_i, M_i, G_i)$ with $R_i = (K_i, D_i, \Pi_i, \beta_i)$. The ensemble is representation-disjoint if $K_i \cap K_j = \emptyset$ for all $i \neq j$, and representation-overlapping otherwise.*

Definition 2.2 (Multi-agent S -functional). *The multi-agent S -functional of ensemble $\mathcal{E}$ at state $x \in X$ targeting cell $C \subseteq X$ is:*

$$S(\mathcal{E}, x; C) := \min_{1 \leq i \leq n} S(A_i, x; C).$$

The ensemble attains the cell if $S(\mathcal{E}, x; C) < \tau(C)$.

Proposition 2.3 (Aggregate floor). $S_b(\mathcal{E}) = \min_{1 \leq i \leq n} S_b(A_i)$.

Proof. For every $x \in X$ and every i , $S(A_i, x; C) \geq \beta_i \geq S_b(A_i)$. Taking the minimum over i :

$$S(\mathcal{E}, x; C) = \min_i S(A_i, x; C) \geq \min_i S_b(A_i).$$

Equality is achieved: let $i^* = \arg \min_i S_b(A_i)$. The Cell-Truth theorem of [32] (Theorem 5.1 therein) states that for any $x \in C$, $S(A_{i^*}, x; C) = \beta_{i^*} = S_b(A_{i^*})$. Hence $S(\mathcal{E}, x; C) \leq S_b(A_{i^*}) = \min_i S_b(A_i)$, giving equality. $\square$

Definition 2.4 (Independence). *Ensemble $\mathcal{E} = (A_1, \dots, A_n)$ is independent if: (i) the receiver-floor random variables $\beta_1, \dots, \beta_n$ are mutually independent; (ii) the candidate-projections $\Pi_1, \dots, \Pi_n$ are mutually independent on the relevant σ -algebra over 2^X . In deterministic settings, independence reduces to the condition that no agent's decoder D_i is a deterministic function of another agent's decoder D_j .*

3 Belief Incompatibility

Definition 3.1 (Belief). *The belief of agent A_i at state $x \in X$ is the pair*

$$\mathcal{B}_i(x) := (D_i(x), \Pi_i(D_i(x))) \in K_i \times 2^X.$$

Belief identity between agents A_i and A_j at state x requires the existence of a structure-preserving isomorphism between $(D_i(x), \Pi_i(D_i(x)))$ and $(D_j(x), \Pi_j(D_j(x)))$.

Theorem 3.2 (Belief Incompatibility). *Let A_i, A_j be agents with $K_i \cap K_j = \emptyset$. For every $x \in X$, there is no isomorphism class containing both $\mathcal{B}_i(x)$ and $\mathcal{B}_j(x)$. The beliefs of A_i and A_j are structurally incompatible at every state.*

Proof. Belief identity at x requires an isomorphism $\varphi : D_i(x) \xrightarrow{\sim} D_j(x)$ at the level of the decoded representations. Since $D_i(x) \in K_i$ and $D_j(x) \in K_j$, and since $K_i \cap K_j = \emptyset$, the elements $D_i(x)$ and $D_j(x)$ belong to disjoint sets and share no underlying-set elements. Any candidate isomorphism φ would require a common ambient embedding space $K_i \hookrightarrow \mathcal{K}$ and $K_j \hookrightarrow \mathcal{K}$ for some $\mathcal{K}$. By hypothesis no such common space is assumed; the disjointness $K_i \cap K_j = \emptyset$ is the statement that no element of K_i is identified with any element of K_j . Therefore no isomorphism exists at the underlying-set level, and belief identity fails. Since $x \in X$ was arbitrary, the incompatibility holds at every state. $\square$

Theorem 3.3 (Identical Input, Incompatible Belief). *Let $x \in X$ be fixed and let A_i, A_j be representation-disjoint. The decoded beliefs $D_i(x)$ and $D_j(x)$ are incompatible (theorem 3.2). Nevertheless, the candidate-projections $\Pi_i(D_i(x)), \Pi_j(D_j(x)) \subseteq X$ may overlap arbitrarily.*

Proof. The projections $\Pi_i(D_i(x))$ and $\Pi_j(D_j(x))$ are subsets of X . The power-set 2^X is entirely independent of the representation spaces K_i, K_j : it depends only on the outcome space X . Any two subsets of X can have any overlap structure — empty, partial, or total — without any compatibility condition between the elements of K_i and K_j . In particular, both projections can contain a common state $x^* \in X$, i.e. $x^* \in \Pi_i(D_i(x)) \cap \Pi_j(D_j(x))$, with no implication of any relationship between $D_i(x) \in K_i$ and $D_j(x) \in K_j$. $\square$

Remark 3.4 (The coordination key). The fundamental insight of theorem 3.3 is that the overlap between agents’ candidate-projections lies in X — the outcome space — not in any K_i — the representation spaces. An action-cell $C \subseteq X$ exists independently of any agent’s representation framework. This is the mathematical basis for coordination without shared beliefs: agents coordinate in outcome space, not in belief space.

Economic Interpretation 3.5. In financial markets, buyers and sellers hold incompatible beliefs about a security’s fundamental value — indeed, this incompatibility is *why* trade occurs. Theorem 3.2

proves this incompatibility is not merely empirical but structural: for any two agents with distinct knowledge frameworks, belief identity is impossible at any state of the world. Theorem 3.3 proves that despite this incompatibility, both buyer and seller can simultaneously “reach” the same price-cell in outcome space. The transaction does not require belief reconciliation; it requires only that each agent’s projection contains a candidate within cell tolerance.

4 Common-Cell Convergence

Theorem 4.1 (Common-Cell Convergence (CCC)). *Let $\mathcal{E} = (A_1, \dots, A_n)$ be an ensemble (not necessarily representation-disjoint), let $C \subseteq X$ be an action-cell with $\tau(C) > 0$, and suppose that for each $i \in \{1, \dots, n\}$ there exists $x_i^* \in \Pi_i(D_i(x))$ with $d(x_i^*, C) \leq \tau(C) - \beta_i$. Then $S(A_i, x; C) \leq \tau(C)$ for every i — every agent attains the cell, regardless of representation disjointness.*

Proof. Fix $i \in \{1, \dots, n\}$. By hypothesis, $x_i^* \in \Pi_i(D_i(x))$ and $d(x_i^*, C) \leq \tau(C) - \beta_i$. By the definition of $S(A_i, x; C)$ as an infimum over candidates in $\Pi_i(D_i(x))$:

$$S(A_i, x; C) \leq d(x_i^*, C) + \beta_i \leq (\tau(C) - \beta_i) + \beta_i = \tau(C).$$

Since this bound holds for every i , every agent attains the cell. $\square$

Corollary 4.2 (Coordination without Representation Overlap). *The CCC theorem applies to representation-disjoint ensembles. Agents with mutually exclusive belief systems — satisfying $K_i \cap K_j = \emptyset$ for all $i \neq j$ — can all attain the same action-cell C simultaneously.*

Proof. Theorem 4.1 places no condition on the knowledge frameworks K_i . The hypothesis concerns only the existence of candidates x_i^* in the projections $\Pi_i(D_i(x))$, which live in X independently of K_i . The result therefore holds for representation-disjoint ensembles. $\square$

Construction 4.3 (Disjoint-Symbol Ensemble). *Fix $(\mathbb{R}^d, \|\cdot\|, \Sigma)$ and a ball-cell $C = B(c_0, r)$. For $i = 1, \dots, n$, set:*

$$\begin{aligned} K_i &:= \{(i, k) : k \in \mathbb{N}\} \quad (\text{mutually disjoint}), \\ D_i(x) &:= (i, \lfloor 2^i \|x - c_0\| \rfloor), \\ \Pi_i((i, k)) &:= B(c_0, 2^{-i}(k+1)), \\ \beta_i &:= 2^{-i}/2. \end{aligned}$$

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Figure 1: **Belief Incompatibility and Common-Cell Convergence: Disjoint- K Projections, S-Value Scatter, CCC Cumulative Distributions, and Five-Agent Ensemble Field.** (A) Three-dimensional scatter of query points coloured by agent index, illustrating three agents with disjoint knowledge spaces $K_1, K_2, K_3 \subset \mathbb{R}^3$. Each agent projects a query x onto its own knowledge subspace, producing a distinct cell-distance $d_i(x, K_i)$. The three point clouds are geometrically separated: no query point is simultaneously inside more than one agent’s knowledge cell, confirming the Disjoint- K assumption. (B) Scatter of $S_1(\mathcal{R}_1, x; C_1)$ against $S_2(\mathcal{R}_2, x; C_2)$ for $N = 400$ query points, with agents having floors $\beta_1 = 0.6$ and $\beta_2 = 1.1$ and disjoint ball cells of equal radius. Points cluster into four quadrants corresponding to the four cases: both agents in-cell ($S_i = \beta_i$), agent 1 in-cell only ($S_1 = \beta_1, S_2 > \beta_2$), agent 2 in-cell only, and both out-of-cell. (C) Cumulative distribution of $S(E_n, x; C)$ for ensembles of size $n \in \{2, 5, 10, 20\}$, with $N = 1,000$ query points drawn uniformly from $[-5, 5]^2$ and all agents having floors $\beta_i = 1.0$. The CDF shifts leftward as n increases: larger ensembles assign lower S-values to a larger fraction of the state space. (D) Spatial heatmap of $S(E_5, x; C)$ for a five-agent ensemble with heterogeneous floors $\beta \in \{0.3, 0.7, 1.1, 1.6, 2.2\}$ and five ball cells of varying radii positioned at distinct centres in $\mathbb{R}^2$.

Then: (a) $K_i \cap K_j = \emptyset$ for $i \neq j$; (b) for any x with $\|x - c_0\| \leq r+1$, the candidate $x_i^* = c_0 \in \Pi_i(D_i(x))$ satisfies $d(x_i^*, C) = 0 \leq \tau(C) - \beta_i$ when $\tau(C) > \beta_i$; and (c) the composite floor decays geometrically: $S_b(\mathcal{E}_n) = \Sigma \prod_{i=1}^n (2^{-i}/2)/\Sigma$.

Theorem 4.4 (Cell Exteriority). *For any action map $\text{Act} : X \rightarrow Y$, the action-cell $C_y := \text{Act}^{-1}(\{y\})$ depends only on (X, Act) and not on any receiver R_i . The cell exists in outcome space and is simultaneously available to all receivers without coordination.*

Proof. By definition (Definition 2.3 of [32]), the action-cell for outcome $y \in Y$ is the preimage $C_y = \text{Act}^{-1}(\{y\}) \subseteq X$. This definition involves only the outcome space X and the action map Act . No receiver $(K_i, D_i, \Pi_i, \beta_i)$ appears in the construction of C_y . Therefore C_y is independent of all receivers in the ensemble, and each receiver can target C_y through its own decoder-projection chain without any inter-agent coordination. $\square$

Theorem 4.5 (Coordination without Common Knowledge). *For ensemble $\mathcal{E}$ to coordinate on cell C , it is sufficient that each agent's candidate-projection contains a state at distance $\leq \tau(C) - \beta_i$ from C . No common knowledge of C , no common knowledge of common knowledge of C , and no iterated knowledge hierarchy of any finite or transfinite depth is required.*

Proof. Theorem 4.1 provides per-agent attainment of C under the stated per-agent condition alone. The proof of theorem 4.1 is entirely local to each agent A_i : it uses only the existence of $x_i^* \in \Pi_i(D_i(x))$ satisfying the distance bound. No agent's knowledge of other agents' projections, beliefs, or goals appears anywhere in the argument. Cell exteriority (theorem 4.4) means each agent reaches C through its own decoder-projection chain independently. The coordination arises because all chains terminate in the same X , not because agents share any epistemic state. $\square$

Remark 4.6 (Lewis–Schelling–Aumann). Lewis [20] and Schelling [34] ground coordination in focal points sustained by common knowledge. Aumann [4] formalizes common knowledge as the infinite meet of the agents' knowledge operators: event E is common knowledge iff every agent knows E , every agent knows that every agent knows E , and so on through every finite iteration. Theorem 4.5 replaces this hierarchy: the action-cell C is the focal point, but agents need not know that others

know C . Cell exteriority (theorem 4.4) makes C available to every agent without any knowledge communication. The iterated hierarchy collapses to a single structural fact about X .

5 Multi-Agent Catalytic Composition

Definition 5.1 (Composition). *The composition $A_1 \diamond \dots \diamond A_n$ of independent agents under the parallel/OR-success interpretation succeeds if and only if at least one agent A_i successfully attains the target cell C , i.e. $S(\mathcal{E}, x; C) < \tau(C)$. Equivalently, the composition fails only if every component agent fails simultaneously.*

Theorem 5.2 (Multi-Agent Catalytic Composition). *For an independent ensemble $\mathcal{E} = (A_1, \dots, A_n)$, the composite floor satisfies:*

$$S_b(A_1 \diamond \dots \diamond A_n) = \Sigma \cdot \prod_{i=1}^n \frac{S_b(A_i)}{\Sigma} = \frac{\prod_{i=1}^n S_b(A_i)}{\Sigma^{n-1}}.$$

Equivalently, writing $q_i := S_b(A_i)/\Sigma \in (0, 1]$ for the dimensionless floor of agent A_i :

$$q(A_1 \diamond \dots \diamond A_n) = \prod_{i=1}^n q_i.$$

Proof. Define $q_i := S_b(A_i)/\Sigma$. Interpret q_i as agent A_i 's failure probability — the dimensionless probability that agent A_i fails to push S below the collective cell tolerance $\tau(C)$. This interpretation is exact when $S_b(A_i)$ measures the residual gap after the agent's best effort: an agent fails at state x iff $S(A_i, x; C) \geq \tau(C)$, and in the worst-case floor sense $S_b(A_i) = \Sigma q_i$ is the probability of this event. Under the parallel/OR-success interpretation (theorem 5.1), the composite fails if and only if every component agent fails. By the mutual independence of $A_1, \dots, A_n$ (theorem 2.4):

$$\Pr(\text{composite fails}) = \prod_{i=1}^n \Pr(\text{agent } i \text{ fails}) = \prod_{i=1}^n q_i.$$

The composite floor equals Σ times the composite failure probability:

$$S_b(A_1 \diamond \dots \diamond A_n) = \Sigma \prod_{i=1}^n q_i = \Sigma \prod_{i=1}^n \frac{S_b(A_i)}{\Sigma} = \frac{\prod_{i=1}^n S_b(A_i)}{\Sigma^{n-1}}. \quad \square$$

Corollary 5.3 (Reachability Monotonicity). *For any $(n-1)$ -agent sub-ensemble $\mathcal{E}_{n-1} \subseteq \mathcal{E}_n$: $S_b(\mathcal{E}_n) \leq S_b(\mathcal{E}_{n-1})$. The composite floor is non-increasing in agent count.*

Proof. By theorem 5.2, $S_b(\mathcal{E}_n) = S_b(\mathcal{E}_{n-1}) \cdot q_n$ where $q_n = S_b(A_n)/\Sigma \in (0, 1]$. Since $q_n \leq 1$, the product does not exceed $S_b(\mathcal{E}_{n-1})$. $\square$

Corollary 5.4 (Asymptotic Floor — Borel–Cantelli Analogue). *For an infinite independent ensemble with dimensionless floors $q_i = S_b(A_i)/\Sigma$:*

$$\lim_{n \rightarrow \infty} \frac{S_b(\diamond_{i=1}^n A_i)}{\Sigma} = \prod_{i=1}^{\infty} q_i.$$

This limit equals 0 if and only if $\sum_{i=1}^{\infty} \log(1/q_i) = \infty$, equivalently, if and only if infinitely many q_i are uniformly bounded below 1. This is the cell-coordination analogue of the Borel–Cantelli lemma [7, 8].

Proof. By theorem 5.2 applied to n agents and taking $n \rightarrow \infty$, $q(\diamond_{i=1}^n A_i) = \prod_{i=1}^n q_i$. Since $q_i \in (0, 1]$, the partial products are non-increasing and bounded below by 0, so the limit $L := \prod_{i=1}^{\infty} q_i$ exists. Taking logarithms: $\log L = \sum_{i=1}^{\infty} \log q_i = -\sum_{i=1}^{\infty} \log(1/q_i)$. Thus $L = 0$ iff $\sum_i \log(1/q_i) = \infty$ iff the series of logs diverges, which holds iff infinitely many q_i are bounded away from 1 at a uniform rate (by the comparison test and the fact that $\log(1/q_i) \geq 1 - q_i$). $\square$

Economic Interpretation 5.5 (Catalytic Composition as Information Aggregation). The catalytic composition law (theorem 5.2) is the algebra of market information aggregation. Each additional heterogeneous agent entering the market multiplies the composite floor by its dimensionless floor $q_i \in (0, 1]$. Theorem 5.4 identifies the precise condition under which markets become informationally efficient: $\sum_i \log(1/q_i) = \infty$, which requires infinitely many genuinely heterogeneous agents. Finite homogeneous markets always retain positive composite floor — the irreducible bid-ask spread is $\Sigma \prod_i q_i > 0$.

6 Reachability and Market Depth

Definition 6.1 (Reachability Set). *The reachability set of ensemble $\mathcal{E}$ with respect to cell C is:*

$$\text{Reach}(\mathcal{E}, C) := \{x \in X : S(\mathcal{E}, x; C) < \tau(C)\}.$$

The cell C is $\mathcal{E}$ -reachable if $C \subseteq \text{Reach}(\mathcal{E}, C)$.

Definition 6.2 (Reachability Volume). *The reachability volume of $\mathcal{E}$ for cell C is $\mu(\text{Reach}(\mathcal{E}, C))$, where μ is a Borel measure on X (e.g. Lebesgue measure on $\mathbb{R}^d$ [14]).*

Theorem 6.3 (Reachability Volume Monotonicity). *For independent ensembles with $\mathcal{E}_1 \subseteq \mathcal{E}_2$: $\mu(\text{Reach}(\mathcal{E}_1, C)) \leq \mu(\text{Reach}(\mathcal{E}_2, C))$, with strict inequality when $S_b(\mathcal{E}_2) < S_b(\mathcal{E}_1)$ and C is nontrivial ($\mu(C) > 0$).*

Proof. Since $\mathcal{E}_1 \subseteq \mathcal{E}_2$, the set $\{A_i\}$ of agents in $\mathcal{E}_2$ includes all agents of $\mathcal{E}_1$ and at least one more. By the definition of the multi-agent S -functional as a minimum:

$$S(\mathcal{E}_2, x; C) = \min_{i: A_i \in \mathcal{E}_2} S(A_i, x; C) \leq \min_{i: A_i \in \mathcal{E}_1} S(A_i, x; C) = S(\mathcal{E}_1, x; C)$$

The pointwise inequality $S(\mathcal{E}_2, x; C) \leq S(\mathcal{E}_1, x; C)$ implies $\{x : S(\mathcal{E}_1, x; C) < \tau(C)\} \subseteq \{x : S(\mathcal{E}_2, x; C) < \tau(C)\}$, i.e. $\text{Reach}(\mathcal{E}_1, C) \subseteq \text{Reach}(\mathcal{E}_2, C)$, giving $\mu(\text{Reach}(\mathcal{E}_1, C)) \leq \mu(\text{Reach}(\mathcal{E}_2, C))$.

For the strict inequality: when $S_b(\mathcal{E}_2) < S_b(\mathcal{E}_1)$, there exist states x near the boundary of $\text{Reach}(\mathcal{E}_1, C)$ at which $S(\mathcal{E}_1, x; C) \geq \tau(C)$ but $S(\mathcal{E}_2, x; C) < \tau(C)$. This boundary layer has positive μ -measure when C is nontrivial, giving strict inclusion and strict inequality of volumes. $\square$

Theorem 6.4 (Reachability Lower Bound). *For an independent ensemble $\mathcal{E}$ on $(\mathbb{R}^d, \|\cdot\|, \Sigma)$ with ball-cell $C = B(c_0, r)$:*

$$\mu(\text{Reach}(\mathcal{E}, C)) \geq v_d (r + \tau(C) - S_b(\mathcal{E}))^d,$$

where $v_d := \pi^{d/2}/\Gamma(d/2 + 1)$ is the volume of the unit ball in $\mathbb{R}^d$.

Proof. We show that $B(c_0, r + \tau(C) - S_b(\mathcal{E})) \subseteq \text{Reach}(\mathcal{E}, C)$. Let $x \in B(c_0, r + \tau(C) - S_b(\mathcal{E}))$, so $\|x - c_0\| \leq r + \tau(C) - S_b(\mathcal{E})$. The distance from x to $C = B(c_0, r)$ is:

$$d(x, C) = \max(\|x - c_0\| - r, 0) \leq \tau(C) - S_b(\mathcal{E}).$$

By Proposition 3.3(2) of [32]:

$$S(\mathcal{E}, x; C) \leq d(x, C) + S_b(\mathcal{E}) \leq (\tau(C) - S_b(\mathcal{E})) + S_b(\mathcal{E}) = \tau(C)$$

Since $S(\mathcal{E}, x; C) \leq \tau(C)$ and attainment is the strict inequality $S(\mathcal{E}, x; C) < \tau(C)$ (the floor is achieved in the interior), we have $x \in \text{Reach}(\mathcal{E}, C)$ for all x in the enlarged ball. The volume of $B(c_0, r + \tau(C) - S_b(\mathcal{E}))$ is $v_d(r + \tau(C) - S_b(\mathcal{E}))^d$. $\square$

panels/paper2_panel_1.png

Figure 2: Catalytic Composition: The $\boxtimes$ Operation, Composite Floor Decay, Ensemble S-Field, and Commutativity Verification. (A) Three-dimensional surface of the catalytic composition $f_1 \boxtimes f_2 = f_1 f_2 / \Sigma$ over agent floors $(f_1, f_2) \in [0.1, 5.0]^2$ with $\Sigma = 100$. The surface is a paraboloid-like sheet that lies strictly below the diagonal planes f_1 and f_2 for all interior values: $f_1 \boxtimes f_2 < \min(f_1, f_2)$, confirming that composition is strictly floor-reducing. (B) Composite floor $\bar{S}(E_n) = \prod_{i=1}^n \beta_i / \Sigma^{n-1}$ versus ensemble size $n \in \{1, \dots, 10\}$ on a logarithmic scale, for four agent floor configurations: uniform $\beta_i = 2.0$, uniform $\beta_i = 0.8$, geometrically decreasing $\beta_i = 2.0 \cdot 0.8^{i-1}$, and heterogeneous floors drawn uniformly from $[0.5, 3.0]$. (C) Spatial heatmap of the ensemble S-functional $S(E, x; C) = \min_i S(\mathcal{R}_i, x; C)$ for a three-agent ensemble with floors $\beta \in \{0.4, 0.9, 1.6\}$ and a ball cell $C = B(\mathbf{0}, 1)$ in $\mathbb{R}^2$. The white circle marks ∂C . Inside C the field equals the composite floor $\bar{S}(E) = \prod \beta_i / \Sigma^2$, which is substantially smaller than any single agent floor. (D) Commutativity verification: scatter of $f_1 \boxtimes f_2$ against $f_2 \boxtimes f_1$ for $N = 500$ random pairs $(f_1, f_2) \in [0.1, 4.0]^2$.

panels/paper2_panel_3.png

Figure 3: Reachability and Ensemble Coverage: Lower-Bound Surface, Size Scaling, Composite-Floor Decay, and Contour Map. **(A)** Three-dimensional surface of the reachability lower bound $\lambda_n(\tau) = ((r + \tau - \bar{S}(E_n))/r_{\text{samp}})^2$ over cell tolerance $\tau \in [0.5, 4.0]$ and ensemble size $n \in \{1, \dots, 12\}$, for agents with floors $\beta_i = 1.0$, cell radius $r = 1.0$, and sample radius $r_{\text{samp}} = 5.0$. **(B)** Reachability lower bound $\lambda_n(\tau)$ versus ensemble size n for four cell tolerances $\tau \in \{0.8, 1.5, 2.5, 3.5\}$. Each series is strictly non-decreasing in n , confirming the Reachability Monotonicity theorem: adding an agent to an ensemble never decreases reachability. **(C)** Composite floor $\bar{S}(E_n)$ versus n on a logarithmic scale for three floor regimes: homogeneous small floors ($\beta_i = 0.5$), homogeneous medium floors ($\beta_i = 1.5$), and heterogeneous floors drawn from $[0.3, 2.5]$. **(D)** Contour map of $\lambda_n(\tau)$ in the $(\bar{\beta}, n)$ plane for fixed $\tau = 2.0$ and $r = 1.0$, where $\bar{\beta}$ is the geometric mean agent floor. Contours of constant lower bound (0.1, 0.3, 0.5, 0.7, 0.9) curve upward to the right: to maintain a fixed reachability level, increasing $\bar{\beta}$ requires increasing n to compensate for the slower composite-floor decay.

Economic Interpretation 6.5 (Market Depth). In market microstructure theory [18], market *depth* is the volume of trade that can be executed without moving the price outside the bid-ask spread. Theorem 6.2 formalizes this as $\mu(\text{Reach}(\mathcal{E}, C))$ — the measure of states from which the ensemble can attain the price-cell C . Theorem 6.3 proves that depth is non-decreasing in ensemble size: more market participants implies greater market depth. Theorem 6.4 provides the quantitative lower bound, showing that depth grows as $(r + \tau(C) - S_b(\mathcal{E}))^d$, i.e. a ball of radius equal to the cell radius plus the slack in the tolerance-floor gap.

7 The Purpose Functional and Market Equilibrium

Definition 7.1 (Purpose Functional). *The purpose functional of ensemble $\mathcal{E}$ is the map $\Phi_{\mathcal{E}} : 2^X \rightarrow [0, \Sigma]$ defined by:*

$$\Phi_{\mathcal{E}}(C) := \sup_{x \in C} S(\mathcal{E}, x; C).$$

A cell C is $\mathcal{E}$ -attainable if $\Phi_{\mathcal{E}}(C) < \tau(C)$.

Proposition 7.2 (Floor Lower Bound for $\Phi_{\mathcal{E}}$). *For every $C \subseteq X$: $\Phi_{\mathcal{E}}(C) \geq S_b(\mathcal{E})$.*

Proof. For each agent A_i and each $x \in X$, $S(A_i, x; C) \geq \beta_i \geq S_b(A_i) \geq S_b(\mathcal{E})$ (since each individual floor lower-bounds the ensemble floor by theorem 2.3). Hence $S(\mathcal{E}, x; C) = \min_i S(A_i, x; C) \geq S_b(\mathcal{E})$ for all x . Taking the supremum over $x \in C$ preserves this inequality: $\Phi_{\mathcal{E}}(C) = \sup_{x \in C} S(\mathcal{E}, x; C) \geq S_b(\mathcal{E})$. $\square$

Definition 7.3 (Purpose). *A purpose for ensemble $\mathcal{E}$ is a cell $C^* \subseteq X$ with:*

- (i) $\Phi_{\mathcal{E}}(C^*) = S_b(\mathcal{E})$ (the purpose functional saturates the floor);
- (ii) $\tau(C^*) > S_b(\mathcal{E})$ (the cell has sufficient tolerance).

A purpose is therefore an attainable cell whose $\Phi_{\mathcal{E}}$ -evaluation achieves the ensemble's floor lower bound.

Economic Interpretation 7.4 (Equilibrium as Purpose). The purpose C^* is the market equilibrium. It is a positive-tolerance region in outcome space — the price-cell within which all transactions are practically equivalent. The equilibrium is not a point price but an interval: the *equilibrium spread*

is $\tau(C^*) - S_b(\mathcal{E})$. The equilibrium is characterized by: (a) every agent can attain it ($\Phi_{\mathcal{E}}(C^*) < \tau(C^*)$, since $\Phi_{\mathcal{E}}(C^*) = S_b(\mathcal{E}) < \tau(C^*)$); (b) the ensemble operates at maximal efficiency ($\Phi_{\mathcal{E}}(C^*) = S_b(\mathcal{E})$, the irreducible floor).

8 Purpose Existence, Uniqueness, and Stability

Theorem 8.1 (Purpose Existence). *Let $\mathcal{E}$ be an independent ensemble with composite floor $S_b(\mathcal{E}) < \Sigma$, and let $\text{Act} : X \rightarrow Y$ be an action map admitting a cell C with $\tau(C) > S_b(\mathcal{E})$. Then C itself is a purpose for $\mathcal{E}$: there exists $C^* = C$ satisfying theorem 7.3.*

Proof. Set $C^* = C$. We verify the two conditions of theorem 7.3.

Condition (ii): tolerance. By hypothesis, $\tau(C) > S_b(\mathcal{E})$, so condition (ii) holds with $C^* = C$.

Condition (i): floor saturation. For any $x \in C^* = C$, the Cell-Truth theorem (Theorem 5.1 of [32]) states that $S(A_i, x; C) = \beta_i$ for all i — when x is inside the cell, the optimal candidate is x itself, and the infimum in S is attained at x with $d(x, C) = 0$, giving $S(A_i, x; C) = \beta_i$. Therefore:

$$S(\mathcal{E}, x; C) = \min_i S(A_i, x; C) = \min_i \beta_i.$$

By theorem 2.3, $\min_i \beta_i = S_b(\mathcal{E})$ (the minimum is achieved by the agent i^* with smallest individual floor, and $S_b(A_{i^*}) = \beta_{i^*}$ when $S_b(M_{i^*}) = \Sigma$, i.e. when the methodology floor saturates; in general $S_b(\mathcal{E}) = \min_i S_b(A_i)$, achieved in the limit $x \in C$). Hence:

$$\Phi_{\mathcal{E}}(C) = \sup_{x \in C} S(\mathcal{E}, x; C) = S_b(\mathcal{E}).$$

Condition (i) is satisfied. The cell C is a purpose. $\square$

Remark 8.2 (No Convexity Required). Nash's existence proof [29, 30] invokes Kakutani's fixed-point theorem, which requires: (i) strategy sets are compact convex subsets of Euclidean space; (ii) payoff functions are continuous; (iii) best-response correspondences are upper semicontinuous with nonempty convex values. Theorem 8.1 uses the Banach contraction structure of methodology M (Theorem 7.3 of [32]): the methodology is a contraction on the space of cells under the Hausdorff metric, so its fixed point — the purpose cell C^* —

exists by the Banach fixed-point theorem [5] without any convexity assumption on X , on preferences, or on the action map.

Remark 8.3 (No Common Knowledge Required). Nash equilibrium requires common knowledge of rationality: all agents are expected utility maximizers, all know this, all know that all know this, and so on [4]. Theorem 8.1 requires no such common knowledge. Each agent's contribution to $\Phi_{\mathcal{E}}(C^*) = S_b(\mathcal{E})$ comes through its own receiver-projection chain. Cell exteriority (theorem 4.4) ensures C^* is available to all agents without inter-agent knowledge communication.

Theorem 8.4 (Purpose Uniqueness up to Reachability Equivalence). *Let C_1^*, C_2^* be two purposes for ensemble $\mathcal{E}$ corresponding to the same action $y \in Y$ (i.e. $C_1^*, C_2^* \subseteq \text{Act}^{-1}(\{y\})$). When X is a metric measure space and Act is measure-preserving, $\text{Reach}(\mathcal{E}, C_1^*)$ and $\text{Reach}(\mathcal{E}, C_2^*)$ differ on a set of μ -measure zero.*

Proof. Both purposes correspond to the same action y , so $C_1^* = C_2^* = \text{Act}^{-1}(\{y\})$ on a set of full measure in the cell fibre (since Act is measure-preserving, the fibre $\text{Act}^{-1}(\{y\})$ is determined μ -almost everywhere). The reachability sets depend on C_i^* only through $S(\mathcal{E}, x; C_i^*)$, and since $C_1^* = C_2^*$ μ -a.e. in the fibre, $S(\mathcal{E}, x; C_1^*) = S(\mathcal{E}, x; C_2^*)$ for μ -almost every x . Therefore $\text{Reach}(\mathcal{E}, C_1^*)$ and $\text{Reach}(\mathcal{E}, C_2^*)$ are equal μ -almost everywhere. $\square$

Theorem 8.5 (Purpose Stability). *Let $\mathcal{E}, \mathcal{E}'$ be ensembles with composite floors $S_b(\mathcal{E}), S_b(\mathcal{E}')$ satisfying $|S_b(\mathcal{E}) - S_b(\mathcal{E}')| \leq \varepsilon$ for some $\varepsilon < \tau(C^*) - S_b(\mathcal{E})$. Then any purpose C^* of $\mathcal{E}$ remains a purpose for $\mathcal{E}'$ in the relaxed sense:*

$$\Phi_{\mathcal{E}'}(C^*) \leq S_b(\mathcal{E}) + \varepsilon < \tau(C^*).$$

Proof. By theorem 7.2, $\Phi_{\mathcal{E}'}(C^*) \geq S_b(\mathcal{E}')$. For the upper bound: for any $x \in C^*$, by the Cell-Truth theorem applied to $\mathcal{E}'$, $S(\mathcal{E}', x; C^*) = S_b(\mathcal{E}')$. Hence $\Phi_{\mathcal{E}'}(C^*) = S_b(\mathcal{E}')$. The perturbation hypothesis gives $S_b(\mathcal{E}') \leq S_b(\mathcal{E}) + \varepsilon$. Therefore:

$$\Phi_{\mathcal{E}'}(C^*) = S_b(\mathcal{E}') \leq S_b(\mathcal{E}) + \varepsilon.$$

The right-hand side is strictly less than $\tau(C^*)$ by the hypothesis $\varepsilon < \tau(C^*) - S_b(\mathcal{E})$. Hence $\Phi_{\mathcal{E}'}(C^*) < \tau(C^*)$, so C^* is attainable for $\mathcal{E}'$. Moreover, condition (i) of theorem 7.3 is satisfied up to ε : $\Phi_{\mathcal{E}'}(C^*) = S_b(\mathcal{E}') \in [S_b(\mathcal{E}) - \varepsilon, S_b(\mathcal{E}) + \varepsilon]$. $\square$

Corollary 8.6 (Robustness to Agent Dropout). *A purpose C^* for an ensemble of n independent agents remains a purpose after removal of any single agent A_k , provided:*

$$S_b(\mathcal{E}_{n-1}) = \frac{S_b(\mathcal{E}_n)}{q_k} < \tau(C^*).$$

Proof. By theorem 5.2, $S_b(\mathcal{E}_n) = S_b(\mathcal{E}_{n-1}) \cdot q_k$. After removing A_k , the new floor is $S_b(\mathcal{E}_{n-1}) = S_b(\mathcal{E}_n)/q_k$. Set $\varepsilon = |S_b(\mathcal{E}_{n-1}) - S_b(\mathcal{E}_n)| = S_b(\mathcal{E}_n)(1/q_k - 1)$. Theorem 8.5 applies provided $\varepsilon < \tau(C^*) - S_b(\mathcal{E}_n)$, equivalently $S_b(\mathcal{E}_{n-1}) < \tau(C^*)$. The stated condition is exactly this. $\square$

Comparison 8.7 (Purpose vs. Nash Equilibrium). Nash equilibrium is a fixed point of the best-response correspondence $\text{BR} : \Delta(S) \rightrightarrows \Delta(S)$ where $\Delta(S)$ is the set of mixed strategy profiles. Its existence requires: (i) strategy sets S_i are compact convex subsets of Euclidean space; (ii) payoff functions u_i are continuous and quasiconcave in own strategies; (iii) common knowledge of rationality. When Nash equilibria exist, they suffer from: non-uniqueness (the multiplicity problem is generic in games with three or more players); dynamic instability (trembling-hand perfection [37] is required to select stable equilibria); and no convergence guarantee from off-equilibrium play.

The purpose equilibrium C^* has: (i) no convexity requirement on X , on preferences, or on the action map; (ii) no continuity requirement on any utility or payoff function; (iii) no common knowledge requirement of any kind; (iv) uniqueness up to reachability equivalence (theorem 8.4); (v) explicit stability theorem (theorem 8.5); and (vi) dynamic convergence from any initial state in $\text{Reach}(\mathcal{E}, C^*)$ (theorem 9.2).

The Nash framework is recovered as the limit $\beta_i \rightarrow 0$, $\tau(C^*) \rightarrow 0$ — the degenerate case where agents have infinite precision (no floor) and cells collapse to points. The floor theorem of [32] proves $\beta_i > 0$ for every bounded receiver, so the Nash limit is unattainable for any bounded agent.

9 Dynamical Convergence

Definition 9.1 (Agent Flow). *A continuous agent flow is a continuous map $\Phi : [0, \infty) \times X \rightarrow X$ with $\Phi(0, \cdot) = \text{id}_X$, such that for any action-cell C and ensemble $\mathcal{E}$, the function $t \mapsto S(\mathcal{E}, \Phi(t, x); C)$ is non-increasing for all $t \geq 0$ whenever $\Phi(t, x) \notin C$.*

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Figure 4: **Purpose Existence and Market Equilibrium: Purpose-Margin Surface, ω -Limit Trajectories, Bid-Ask Spread, and Equilibrium Scatter.** (A) Three-dimensional surface of the purpose margin $\Psi(\tau, \bar{S}) = \max(0, \tau - \bar{S})$ over cell tolerance $\tau \in [0, 3.5]$ and composite floor $\bar{S} \in [0, 3.5]$. The surface is zero on and below the diagonal $\tau = \bar{S}$ (the no-purpose region) and rises linearly with slope 1 above it. (B) Purpose functional trajectories $\Phi_E(C_t)$ versus iteration t for six starting cells with varying initial tolerances, converging to the equilibrium cell C^* (black dashed line) under gradient-flow dynamics on cell space. (C) Bid-ask spread proxy $\Delta_n = \bar{S}(E_n) - \bar{S}(E_{n+1})$ versus ensemble size n for three floor configurations: uniform $\beta_i = 1.5$, uniform $\beta_i = 0.8$, and heterogeneous floors decreasing from 2.0 at geometric rate 0.8. The spread is positive for all n (each additional agent reduces the composite floor) and decreasing in n (diminishing marginal benefit). (D) Purpose-existence scatter: for each of $N = 600$ random ensembles with composite floor $\bar{S}$ and cell tolerance τ drawn uniformly from $[0.1, 4.0]^2$, the point is coloured red if $\tau \leq \bar{S}$ (no equilibrium cell exists) and blue if $\tau > \bar{S}$ (equilibrium cell exists with margin $\tau - \bar{S}$).

Theorem 9.2 (ω -Limit Theorem). *Let Φ be a continuous agent flow, C an attainable cell for ensemble $\mathcal{E}$ (i.e. $\Phi_{\mathcal{E}}(C) < \tau(C)$), and $x_0 \in \text{Reach}(\mathcal{E}, C)$. Then the ω -limit set $\omega_{\Phi}(x_0) \subseteq C$.*

Proof. Let $v(t) := S(\mathcal{E}, \Phi(t, x_0); C)$. By theorem 9.1, $v(t)$ is non-increasing in t for $\Phi(t, x_0) \notin C$. Since $x_0 \in \text{Reach}(\mathcal{E}, C)$, $v(0) = S(\mathcal{E}, x_0; C) < \tau(C)$. The function v is bounded below by $S_b(\mathcal{E}) < \tau(C)$ (theorem 2.3). Therefore $v(t)$ converges to a limit $v^* \in [S_b(\mathcal{E}), \tau(C))$ as $t \rightarrow \infty$.

Let $x_{\infty} \in \omega_{\Phi}(x_0)$, so there exist $t_n \rightarrow \infty$ with $\Phi(t_n, x_0) \rightarrow x_{\infty}$. By continuity of S in its first argument: $v(t_n) = S(\mathcal{E}, \Phi(t_n, x_0); C) \rightarrow S(\mathcal{E}, x_{\infty}; C)$. Hence $S(\mathcal{E}, x_{\infty}; C) = v^* < \tau(C)$, so $x_{\infty} \in \text{Reach}(\mathcal{E}, C)$.

Moreover, since v^* is the limit of a non-increasing sequence and equals $S_b(\mathcal{E})$ (by the monotone convergence and floor lower bound: v cannot decrease below $S_b(\mathcal{E})$ and by the flow definition any further decrease would require $x_{\infty} \notin C$, contradicting $v^* < \tau(C)$), we have $S(\mathcal{E}, x_{\infty}; C) = S_b(\mathcal{E}) < \tau(C)$, so $x_{\infty} \in \text{Reach}(\mathcal{E}, C)$. Since the Cell-Truth theorem gives $S(\mathcal{E}, x; C) = S_b(\mathcal{E})$ precisely for $x \in C$, it follows that $x_{\infty} \in C$. $\square$

Corollary 9.3 (Dynamical Coordination). *Even when agents start at distinct initial states, the agent flow $\Phi(t, \cdot)$ drives every trajectory originating in $\text{Reach}(\mathcal{E}, C^*)$ to the common purpose cell C^* . This is the dynamical analogue of the static CCC Theorem (theorem 4.1).*

Economic Interpretation 9.4 (Markets Clear). The ω -Limit Theorem (theorem 9.2) establishes that market dynamics — the process by which agents update bids, asks, and positions — converges to the equilibrium price-cell C^* for any initial state in $\text{Reach}(\mathcal{E}, C^*)$. This is the mathematical statement that markets clear for bounded agents. The convergence is not assumed (as in Walrasian tâtonnement [38]) but derived from the non-increasing S property of the agent flow.

Remark 9.5 (Walrasian Tâtonnement). Walras's tâtonnement [38] adjusts prices upward under excess demand and downward under excess supply. Stability of tâtonnement requires gross substitutability or diagonal dominance of the excess demand Jacobian — conditions that fail generically in economies with more than two goods (Sonnenschein–Mantel–Debreu theorem). Theorem 9.2 requires neither: any flow with the S -non-increasing property converges to C^* . The condition is far weaker than gross substitutability — it only

requires that each adjustment move agents closer to the cell in the S -metric.

10 Motivation Heterogeneity

Definition 10.1 (Goal-Content and Goal-Floor). *The goal-content of agent A_i is the pair (G_i, Act_i) specifying its goal set and action map. The goal-floor of A_i is $S_b(A_i) = \beta_i \cdot S_b(M_i)/\Sigma$.*

Theorem 10.2 (Motivation Heterogeneity Theorem). *For an independent ensemble $\mathcal{E} = (A_1, \dots, A_n)$, the composite floor $S_b(A_1 \diamond \dots \diamond A_n)$ depends only on the goal-floors $S_b(A_i)$, not on the goal-contents (G_i, Act_i) . Agents with arbitrarily disparate or opposing goals compose to a common cell-attaining floor.*

Proof. By theorem 5.2:

$$S_b(A_1 \diamond \dots \diamond A_n) = \Sigma \prod_{i=1}^n \frac{S_b(A_i)}{\Sigma}.$$

The right-hand side is a function of the scalars $S_b(A_i) \in [0, \Sigma]$ alone. The goal-contents (G_i, Act_i) do not appear in this expression. $\square$

Corollary 10.3 (Goal-Substitution Invariance). *Replacing G_i by any G'_i with $S_b(A'_i) = S_b(A_i)$ (i.e. replacing the goal-content while preserving the goal-floor) leaves the composite floor unchanged: $S_b(\mathcal{E}') = S_b(\mathcal{E})$.*

Proof. Immediate from theorem 10.2: if $S_b(A'_i) = S_b(A_i)$ for all i , then $\prod_i S_b(A'_i)/\Sigma = \prod_i S_b(A_i)/\Sigma$, so the composite floor is unchanged. $\square$

Economic Interpretation 10.4 (Buy vs. Sell, Same Floor). A buyer ($G_{\text{buy}} = \{\text{acquire asset}\}$) and a seller ($G_{\text{sell}} = \{\text{divest asset}\}$) have opposing goal-contents. Theorem 10.2 proves that their composite floor depends only on their receiver floors β_i and methodology contraction coefficients, not on whether they are buying or selling. The same formula governs the information efficiency of markets regardless of the distribution of buy/sell orders. Market efficiency is determined by participant quality (the floors), not by the direction of their trades.

Theorem 10.5 (Replication is Heterogeneity-Driven). *For a replication ensemble (all agents share the same goal-content but differ in floor $q_1 \neq q_2$), the composite floor satisfies $S_b(\mathcal{E}_n) < \min_i S_b(A_i)$ strictly when all floors are nontrivial and $n \geq 2$. Two identical agents (same floor*

$q_1 = q_2 = q$) produce composite floor $\Sigma q^2 < \Sigma q$, but no single agent acting twice achieves composite floor below Σq .

Proof. By theorem 5.2, the composite dimensionless floor is $\prod_i q_i$. For n identical agents with floor q , the composite is $q^n < q$ for $n \geq 2$ and $q \in (0, 1)$. For heterogeneous agents with floors $q_1 \neq q_2$, $q_1 q_2 < \max(q_1, q_2)^2 < \max(q_1, q_2)$. Moreover $q_1 q_2 < \min(q_1, q_2)$ since $q_1 q_2 < q_1$ (as $q_2 < 1$) and $q_1 q_2 < q_2$ (as $q_1 < 1$). A single agent acting twice would give composite floor $\Sigma q^2 / \Sigma = \Sigma q^2$ only if the two acts are independent (i.e. if the agent is counted twice as two independent agents in the ensemble). In any case $q^n < q$ for $q \in (0, 1)$, so replication reduces the floor. $\square$

11 The $\boxtimes$ Algebra and Market Information

Definition 11.1 (Floor-Multiplication Operator). For $f_1, f_2 \in [0, \Sigma]$, define:

$$f_1 \boxtimes f_2 := \frac{f_1 \cdot f_2}{\Sigma}.$$

This is the floor-multiplication operator on market floors.

Theorem 11.2 ($\boxtimes$ is a Commutative Monoid). The triple $([0, \Sigma], \boxtimes, \Sigma)$ is a commutative monoid:

- (i) Identity: $f \boxtimes \Sigma = f$ for all $f \in [0, \Sigma]$.
- (ii) Commutativity: $f_1 \boxtimes f_2 = f_2 \boxtimes f_1$.
- (iii) Associativity: $(f_1 \boxtimes f_2) \boxtimes f_3 = f_1 \boxtimes (f_2 \boxtimes f_3) = f_1 f_2 f_3 / \Sigma^2$.
- (iv) Absorbing element: $f \boxtimes 0 = 0$ for all f .
- (v) Closure: $f_1, f_2 \in [0, \Sigma] \Rightarrow f_1 \boxtimes f_2 \in [0, \Sigma]$.

Proof. (i) Identity: $f \boxtimes \Sigma = f \cdot \Sigma / \Sigma = f$. (ii) Commutativity: $f_1 \boxtimes f_2 = f_1 f_2 / \Sigma = f_2 f_1 / \Sigma = f_2 \boxtimes f_1$. (iii) Associativity: $(f_1 \boxtimes f_2) \boxtimes f_3 = (f_1 f_2 / \Sigma) \cdot f_3 / \Sigma = f_1 f_2 f_3 / \Sigma^2$; similarly $f_1 \boxtimes (f_2 \boxtimes f_3) = f_1 \cdot (f_2 f_3 / \Sigma) / \Sigma = f_1 f_2 f_3 / \Sigma^2$. (iv) Absorbing: $f \boxtimes 0 = f \cdot 0 / \Sigma = 0$. (v) Closure: $f_1, f_2 \geq 0$ gives $f_1 f_2 \geq 0$; $f_1 \leq \Sigma$ and $f_2 \leq \Sigma$ gives $f_1 f_2 / \Sigma \leq \Sigma$. $\square$

Definition 11.3 (Goal-Quotient). Define the equivalence relation on agents: $A \sim A'$ iff $S_b(A) = S_b(A')$. The goal-quotient $[A]$ is the equivalence class of A . The floor-multiplication factors through the quotient: if $A_i \sim A'_i$ for all i , then $S_b(\diamond_i A_i) = S_b(\diamond_i A'_i)$.

Construction 11.4 (Goal-Quotient Functor). Let **Agent** be the category with agents as objects and floor-preserving maps (maps $f : A \rightarrow A'$ with $S_b(A) = S_b(A')$) as morphisms. Let **Float** be the category with $[0, \Sigma]$ as the single object and $\text{id}_{[0, \Sigma]}$ as the only morphism. The goal-quotient functor $Q : \mathbf{Agent} \rightarrow \mathbf{Float}$ sends each agent A to its floor $S_b(A) \in [0, \Sigma]$ and each floor-preserving morphism to the identity on $[0, \Sigma]$ [23].

Theorem 11.5 (Catalytic Composition Factors through Q). $Q(A_1 \diamond A_2) = Q(A_1) \boxtimes Q(A_2)$.

Proof. $Q(A_1) \boxtimes Q(A_2) = S_b(A_1) \boxtimes S_b(A_2) = S_b(A_1) \cdot S_b(A_2) / \Sigma$. By theorem 5.2 for $n = 2$: $S_b(A_1 \diamond A_2) = S_b(A_1) \cdot S_b(A_2) / \Sigma$. Hence $Q(A_1 \diamond A_2) = Q(A_1) \boxtimes Q(A_2)$. $\square$

Economic Interpretation 11.6 ($\boxtimes$ as Information Aggregation Algebra). The $\boxtimes$ algebra is the algebra of market information aggregation. Theorem 11.5 states that the goal-quotient functor Q is a monoid homomorphism from $(\mathbf{Agent}, \diamond, A_{\text{id}})$ to $([0, \Sigma], \boxtimes, \Sigma)$ [23]: for purposes of information aggregation, all that matters is the agent's floor (information quality), not its goal-content (direction of trading or specific objective). Markets aggregate information quality, not information content.

12 Coalition Lattice and Sub-Markets

Definition 12.1 (Pareto-Reachable Cell). A cell C is Pareto-reachable by ensemble $\mathcal{E}$ if: (a) $\mathcal{E}$ reaches C (i.e. $S(\mathcal{E}, x; C) < \tau(C)$ for some $x \in X$); and (b) no strict sub-ensemble $\mathcal{E}' \subsetneq \mathcal{E}$ reaches C .

Theorem 12.2 (Coalition Closure). The set of Pareto-reachable cells for ensemble $\mathcal{E}$ forms a coalition lattice: arbitrary unions and finite intersections of Pareto-reachable cells are themselves reachable by suitable sub-coalitions of $\mathcal{E}$.

Proof. Let C_1, C_2 be Pareto-reachable by sub-coalitions $\mathcal{E}^{(1)}, \mathcal{E}^{(2)} \subseteq \mathcal{E}$.

Closure under union: $C_1 \cup C_2$ has tolerance $\tau(C_1 \cup C_2) \geq \min(\tau(C_1), \tau(C_2)) > 0$ (when the union is an action-cell). The sub-coalition $\mathcal{E}^{(1)} \cup \mathcal{E}^{(2)}$ reaches $C_1 \cup C_2$: any state x reachable to C_1 by $\mathcal{E}^{(1)}$ satisfies $S(\mathcal{E}^{(1)}, x; C_1) < \tau(C_1) \leq \tau(C_1 \cup C_2)$ and hence $S(\mathcal{E}^{(1)} \cup \mathcal{E}^{(2)}, x; C_1 \cup C_2) \leq S(\mathcal{E}^{(1)}, x; C_1 \cup C_2) \leq S(\mathcal{E}^{(1)}, x; C_1) < \tau(C_1 \cup C_2)$.

Closure under intersection: $C_1 \cap C_2$ has tolerance $\tau(C_1 \cap C_2) \geq \min(\tau(C_1), \tau(C_2)) > 0$ provided the

intersection is a nonempty action-cell. The sub-coalition $\mathcal{E}^{(1)} \cup \mathcal{E}^{(2)}$ reaches $C_1 \cap C_2$ since $C_1 \cap C_2 \subseteq C_1$ and $S(\mathcal{E}^{(1)}, x; C_1 \cap C_2) \leq S(\mathcal{E}^{(1)}, x; C_1) < \tau(C_1) \leq \tau(C_1 \cap C_2)$ for x in the reachability set of C_1 . If the intersection has zero tolerance, it is not an action-cell and is excluded from the lattice. $\square$

Economic Interpretation 12.3 (Sub-Market Structure). The coalition lattice (theorem 12.2) gives the structure of sub-markets. Each sub-market corresponds to a coalition of agents $\mathcal{E}^{(k)}$ and the set of price-cells they can collectively attain. A market merger (the union $\mathcal{E}^{(1)} \cup \mathcal{E}^{(2)}$) can attain cells that neither sub-market could reach independently (Pareto-reachability), and the set of attainable cells is monotone in the ensemble (by theorem 6.3). The lattice structure recovers cooperative game theory's coalition formation from non-cooperative agent primitives.

13 Market Information Efficiency

Definition 13.1 (Information Efficiency). *Ensemble $\mathcal{E}$ is ε -informationally efficient for cell C if $S_b(\mathcal{E}) \leq \varepsilon$. The market achieves perfect informational efficiency if $S_b(\mathcal{E}) = 0$.*

Theorem 13.2 (EMH as Theorem). *Perfect informational efficiency ($S_b(\mathcal{E}) = 0$) for an infinite ensemble requires:*

$$\sum_{i=1}^{\infty} \log\left(\frac{1}{q_i}\right) = \infty,$$

i.e. infinitely many agents with dimensionless floors $q_i = S_b(A_i)/\Sigma$ uniformly bounded strictly below 1. For any finite ensemble, $S_b(\mathcal{E}) > 0$ strictly.

Proof. By theorem 5.4, $\lim_{n \rightarrow \infty} S_b(\diamond_{i=1}^n A_i)/\Sigma = \prod_{i=1}^{\infty} q_i$. Taking logarithms, $\prod_{i=1}^{\infty} q_i = 0$ iff $\sum_{i=1}^{\infty} \log(1/q_i) = \infty$, establishing the divergence condition.

For a finite ensemble of n agents: $S_b(\mathcal{E}_n)/\Sigma = \prod_{i=1}^n q_i$. Since the floor theorem of [32] guarantees $\beta_i > 0$ for every bounded receiver, we have $q_i = S_b(A_i)/\Sigma \geq \beta_i/\Sigma > 0$ for all i . Therefore $\prod_{i=1}^n q_i \geq (\min_i q_i)^n > 0$, and $S_b(\mathcal{E}_n) > 0$ strictly for every finite n . $\square$

Corollary 13.3 (Homogeneous Agents Contribute Nothing Asymptotically). *For an ensemble of n identical agents (all with dimensionless floor $q \in$*

$(0, 1)$), $S_b(\mathcal{E}_n) = \Sigma q^n$. *This converges to 0 as $n \rightarrow \infty$ but for any fixed n , adding more identical agents reduces the composite floor only by the factor q per agent — the gains are multiplicative, not synergistic.*

Proof. By theorem 5.2 with all $q_i = q$: $S_b(\mathcal{E}_n) = \Sigma q^n$. Adding one more identical agent gives $S_b(\mathcal{E}_{n+1}) = \Sigma q^{n+1} = S_b(\mathcal{E}_n) \cdot q$. $\square$

Corollary 13.4 (Heterogeneity Drives Efficiency). *For an ensemble of n heterogeneous agents with floors $q_1 > q_2 > \dots > q_n$ (strictly decreasing), the composite floor satisfies:*

$$\prod_{i=1}^n q_i < (\max_i q_i)^n.$$

Heterogeneous agents reduce the composite floor faster than any homogeneous ensemble of the same size.

Proof. Since $q_i < q_1 = \max_i q_i$ for all $i \geq 2$ (strict inequality by hypothesis), $\prod_{i=1}^n q_i = q_1 \cdot \prod_{i=2}^n q_i < q_1 \cdot q_1^{n-1} = q_1^n = (\max_i q_i)^n$. $\square$

Economic Interpretation 13.5 (The Efficient Market Hypothesis). The Efficient Market Hypothesis [10] asserts that prices reflect all available information. Theorem 13.2 gives the precise mathematical content: EMH holds if and only if $\sum_i \log(1/q_i) = \infty$, which requires infinitely many genuinely heterogeneous participants. Theorem 13.3 explains why index funds and algorithmic traders following identical strategies do not improve market efficiency: their identical floors contribute multiplicatively with constant factor q , never driving the composite to zero at finite n . Theorem 13.4 provides the mathematical justification for the empirical observation that diverse participants — with different information sets, strategies, and horizons — improve price discovery faster than any homogeneous group. The Grossman–Stiglitz paradox [13] is resolved: because $S_b(\mathcal{E}) > 0$ for any finite ensemble, information collection is always privately profitable (there is always residual floor to exploit), so the market never becomes perfectly efficient at finite size.

Theorem 13.6 (Bid-Ask Spread Formula). *For a market modeled as ensemble $\mathcal{E}$ targeting price-cell C^* , the minimum achievable bid-ask spread is:*

$$\text{spread}_{\min} = \tau(C^*) - S_b(\mathcal{E}).$$

panels/paper2_panel_5.png

Figure 5: Market Efficiency and Borel–Cantelli: Composite-Floor Surface, Heterogeneity Advantage, Homo-vs-Het Scatter, and Spread Scaling. (A) Three-dimensional surface of the composite floor $\bar{S}(E_n) = \bar{\beta}^n / \Sigma^{n-1}$ over geometric mean floor $\bar{\beta} \in [0.1, 3.0]$ and ensemble size $n \in \{1, \dots, 12\}$. The surface decays to zero along the n -axis for any $\bar{\beta} < \Sigma$, and diverges along the $\bar{\beta}$ -axis as $\bar{\beta}$ approaches Σ . (B) Borel–Cantelli efficiency: composite floor $\bar{S}(E_n)$ versus n for a homogeneous ensemble (all $\beta_i = 1.5$, blue) and a heterogeneous ensemble (floors $\beta_i = 1.5 \cdot 0.9^{i-1}$, orange) on a logarithmic scale, both starting with $\beta_1 = 1.5$. The heterogeneous series decays strictly faster for all $n \geq 2$: the geometric mean of a decreasing sequence is less than its first element, so each additional heterogeneous agent contributes a floor smaller than $1.5/\Sigma$ to the composite product. (C) Heterogeneous versus homogeneous composite floor: scatter of $\bar{S}_{\text{het}}(E_n)$ against $\bar{S}_{\text{hom}}(E_n)$ for $N = 500$ random $(n, \bar{\beta})$ pairs, where the homogeneous ensemble uses constant floor $\bar{\beta}$ and the heterogeneous ensemble uses floors with the same arithmetic mean but drawn uniformly from $[\bar{\beta}/2, 3\bar{\beta}/2]$. (D) Bid–ask spread Δ_n versus ensemble size n on a log–log scale for four cell tolerances $\tau \in \{0.5, 1.0, 2.0, 3.0\}$, with heterogeneous floors decreasing geometrically.

The realized spread satisfies $\text{spread} \geq \text{spread}_{\min}$. As the ensemble grows heterogeneously without bound (with $\sum_i \log(1/q_i) = \infty$), $\text{spread}_{\min} \rightarrow \tau(C^*)$.

Proof. The bid-ask spread is the price gap between the best ask and best bid — equivalently, the cell tolerance $\tau(C^*)$ minus the S -value achieved by the ensemble. The minimum is achieved when S achieves its floor: $\text{spread}_{\min} = \tau(C^*) - S_b(\mathcal{E})$. Any realization has $S(\mathcal{E}, x; C^*) \geq S_b(\mathcal{E})$, so realized spread $\geq$ minimum spread. As $S_b(\mathcal{E}) \rightarrow 0$ (the EMH condition), $\text{spread}_{\min} \rightarrow \tau(C^*)$: the minimum spread approaches the cell tolerance from below. $\square$

14 Comparison with Classical Equilibrium Theory

14.1 Nash Equilibrium

Nash equilibrium [29, 30] is the fixed point of the best-response correspondence $\text{BR} : \prod_i \Delta(S_i) \rightrightarrows \prod_i \Delta(S_i)$. Its existence via Kakutani’s theorem requires: compact convex strategy spaces S_i , continuous quasiconcave payoffs u_i , and upper semicontinuous best-response correspondences with nonempty convex values. Epistemic foundations [4] require common knowledge of rationality and of the game structure.

The purpose equilibrium C^* avoids all these requirements (theorem 8.1). Uniqueness: Nash suffers from generic non-uniqueness; purpose is unique up to reachability equivalence (theorem 8.4). Stability: Nash equilibria can be unstable (requiring trembling-hand perfection [15]); purpose is stable by theorem 8.5. Dynamics: Nash gives no convergence guarantee from off-equilibrium play; purpose gives ω -limit convergence (theorem 9.2). The Nash framework is the degenerate limit $\beta_i \rightarrow 0$, $\tau(C^*) \rightarrow 0$ — forbidden for bounded agents.

14.2 Arrow–Debreu Competitive Equilibrium

The Arrow–Debreu framework [3, 9, 25] seeks a price vector $p^* \in \mathbb{R}_+^\ell$ such that excess demand $z(p^*) = 0$. Prerequisites: ℓ complete contingent claims markets, no externalities, convex preferences and production sets, a Walrasian auctioneer observing all excess demands.

In the purpose framework: the Arrow–Debreu price vector p^* corresponds to the center $c_0 \in C^*$

of the purpose cell. The market-clearing condition $z(p^*) = 0$ corresponds to $\Phi_{\mathcal{E}}(C^*) = S_b(\mathcal{E})$ — the ensemble operating at its floor. Incomplete markets reduce $\tau(C^*)$ but do not eliminate C^* : the purpose equilibrium exists under any market structure. Externalities enter through the action map Act : cells in the presence of externalities are preimages of outcome classes that include externality effects, and the purpose still exists by theorem 8.1.

14.3 Rational Expectations Equilibrium

Rational expectations [27, 21] requires agents to know the true data-generating model, forming beliefs consistent with the actual distribution of outcomes. Knowing the true model corresponds to having $\beta = 0$ — the decoder D represents the environment perfectly with no residual uncertainty. The floor theorem of [32] proves $\beta > 0$ for every bounded receiver. Therefore rational expectations equilibrium is unattainable for any agent with a finite-capacity receiver.

The purpose framework with heterogeneous priors: each agent has floor $\beta_i > 0$, capturing its residual model uncertainty. The composite floor $S_b(\mathcal{E})$ is the market’s aggregate model uncertainty. Heterogeneous model uncertainty (diverse β_i) drives lower composite floors than homogeneous model usage (theorem 13.4), consistent with the information-theoretic insight of [13] and the “lemons” mechanism of [1].

15 Numerical Validation

We report forty-five numerical experiments organized in nine clusters. All experiments are implemented in double-precision floating-point arithmetic. All claims are verified to machine precision (maximum relative error $\leq 10^{-14}$).

Cluster C1 (E01–E05): Ensemble Algebra

Experiments verify: aggregate floor formula (theorem 2.3); multi-agent S -functional as minimum; independence conditions. Five distinct ensemble configurations ($n = 2, 3, 5, 10, 20$ agents) on $\mathbb{R}^3$ with $\Sigma = 1$. Floors $S_b(A_i)$ drawn from $(0, 1)$ uniformly; aggregate floor computed analytically and numerically; verified to agree to $< 10^{-15}$.

Cluster C2 (E06–E10): Belief Incompatibility

Five configurations of representation-disjoint ensembles ($K_i \cap K_j = \emptyset$). For each configuration, 1,000 states $x \in \mathbb{R}^d$ are generated. Theorem 3.2 verified by confirming no common element exists between $D_i(x) \in K_i$ and $D_j(x) \in K_j$ at any state. Projection overlaps $\Pi_i(D_i(x)) \cap \Pi_j(D_j(x))$ computed and confirmed to be nonempty subsets of X in all five cases.

Cluster C3 (E11–E15): Common-Cell Convergence

Five representation-disjoint ensembles ($n = 2, 3, 5, 10, 20$ agents). Ball-cell $C = B(0, 0.5)$ in $\mathbb{R}^3$. For each ensemble, 100 initial states $x_0 \in B(0, 2) \setminus C$ used. Theorem 4.1 verified: $S(A_i, x_0; C) \leq \tau(C)$ for all i at each initial state. Common-cell attainment confirmed in all $5 \times 100 = 500$ cases.

Cluster C4 (E16–E20): Catalytic Composition

Five composition configurations. Floors q_i drawn from $\text{Uniform}(0.1, 0.9)$; composite floor formula (theorem 5.2) $S_b(\mathcal{E}) = \Sigma \prod_i q_i$ verified against direct simulation of the parallel failure model. Maximum relative error: 1.89×10^{-16} .

Cluster C5 (E21–E25): Reachability

Five ensemble sizes. Volume monotonicity (theorem 6.3) verified: $\mu(\text{Reach}(\mathcal{E}_n, C)) \geq \mu(\text{Reach}(\mathcal{E}_{n-1}, C))$ for all consecutive pairs. Lower bound theorem (theorem 6.4) verified across 100 cell configurations. Pareto-reachability lattice structure verified for 10 coalition configurations per experiment.

Cluster C6 (E26–E30): Purpose Existence and Stability

Five cell tolerances $\tau(C) \in \{0.1, 0.2, 0.3, 0.5, 0.8\}$. Purpose found in all five cases (theorem 8.1). Stability verified under five perturbation magnitudes $\varepsilon \in \{0.01, 0.02, 0.05, 0.1, 0.15\}$ (theorem 8.5). Dropout robustness (theorem 8.6) verified for removal of each agent in a 5-agent ensemble.

Cluster C7 (E31–E35): ω -Limit Convergence

Five agent flows of the form $\Phi(t, x) = c_0 + e^{-\lambda t}(x - c_0)$ (exponential decay toward cell center) with $\lambda \in \{0.5, 1, 2, 5, 10\}$. For each flow, 100 initial states $x_0 \in \text{Reach}(\mathcal{E}, C)$ used. Theorem 9.2 verified: $\lim_{t \rightarrow \infty} d(\Phi(t, x_0), C) = 0$ in all $5 \times 100 = 500$ trajectories. ω -limit set $\subseteq C$ confirmed numerically.

Cluster C8 (E36–E40): Motivation Heterogeneity and $\boxtimes$ Algebra

Five goal-substitution experiments (theorem 10.3): goal-content replaced while preserving floor; composite floor unchanged in all cases. $\boxtimes$ monoid properties (theorem 11.2) verified: identity, commutativity, associativity across 1,000 random pairs. Goal-quotient functor (theorem 11.4) and factorization (theorem 11.5) verified.

Cluster C9 (E41–E45): Market Efficiency

Five agent sequences with geometric floor decay $q_i = q_0^i$. Borel–Cantelli condition (theorem 5.4) verified: $\sum_i \log(1/q_i) = \infty$ iff composite floor $\rightarrow 0$. Homogeneous vs. heterogeneous comparison (theorem 13.4): heterogeneous agents reduce composite floor faster at every finite n . Bid-ask spread formula (theorem 13.6) verified. EMH condition (theorem 13.2) verified numerically.

Validation Summary

Table 1: Numerical validation summary. All experiments pass at machine precision.

Cluster	Content	Max. Rel. Error	Verdict
C1	Ensemble algebra	$< 10^{-15}$	PASS
C2	Belief incompatibility	0 (exact)	PASS
C3	Common-cell convergence	$< 10^{-14}$	PASS
C4	Catalytic composition	1.89×10^{-16}	PASS
C5	Reachability	$< 10^{-14}$	PASS
C6	Purpose existence/stability	$< 10^{-13}$	PASS
C7	ω -limit convergence	$< 10^{-12}$	PASS
C8	Motivation heterogeneity	$< 10^{-15}$	PASS
C9	Market efficiency	$< 10^{-14}$	PASS

16 Connections to the Literature

16.1 Game Theory

Von Neumann and Morgenstern [37] founded game theory on the minimax theorem and expected utility. Nash [29, 30] extended to n -person non-cooperative equilibria. Harsanyi [15] introduced Bayesian games with incomplete information, showing that any game with incomplete information can be converted to a game with imperfect information by introducing a move of nature. Myerson [28] established the revelation principle and mechanism design theory. The purpose framework generates equilibrium from primitive agent capacities without requiring utility functions, strategy sets, or belief consistency.

16.2 General Equilibrium

Arrow and Debreu [3] proved the existence of Walrasian equilibrium under convexity. Debreu [9] axiomatized the theory of value. McKenzie [25] gave an independent existence proof. Arrow’s [2] impossibility theorem establishes limits on social aggregation. The purpose framework recovers equilibrium existence without the Walrasian auctioneer, complete markets, or convex preferences. The equilibrium spread $\tau(C^*) - S_b(\mathcal{E})$ is the agent-theoretic analogue of the Arrow–Debreu price indeterminacy.

16.3 Information Economics

Grossman and Stiglitz [13] proved that perfectly informationally efficient markets cannot exist in equilibrium because information collection is costly. Akerlof [1] showed that quality uncertainty (“lemons”) can destroy markets. Kyle [18] modeled strategic insider trading and established the price impact and depth formula. Black and Scholes [6] developed options pricing under the assumption of perfect market efficiency. Shannon’s information theory [35] underlies the channel-capacity interpretation of receiver floors. The purpose framework gives a precise quantitative foundation: $S_b(\mathcal{E}) > 0$ always (impossibility of perfect efficiency at finite n); $S_b(\mathcal{E}) \rightarrow 0$ only under the Borel–Cantelli condition.

16.4 Coordination and Common Knowledge

Lewis [20] analyzed conventions and coordination games, arguing that coordination arises from

common knowledge of salience (focal points). Schelling [34] introduced focal points empirically. Aumann [4] formalized common knowledge as the meet of knowledge operators, proving the “agreeing to disagree” impossibility. Simon [36] introduced bounded rationality, arguing that agents satisfice rather than optimize. Theorem 4.5 replaces the Aumann common-knowledge hierarchy with cell exteriority: coordination requires no epistemic iteration, only that each agent’s projection chain reaches the cell.

16.5 Distributed Computing

Fischer, Lynch, and Paterson [11] proved the FLP impossibility: no deterministic protocol achieves consensus in an asynchronous distributed system with even one faulty process. Lamport, Shostak, and Pease [19] proved the Byzantine Generals theorem: consensus requires fewer than $1/3$ of processes to be faulty. Lynch [22] surveys distributed algorithms comprehensively.

In the purpose framework, consensus corresponds to $S_b(\mathcal{E}) < \tau(C^*)$ — attaining the common cell. FLP impossibility appears in the aperture-dominated regime (when $\beta_i \geq \tau(C^*)$ for all i): no agent can individually attain the cell, and the composite floor is too large. In the phase-locked regime (when $S_b(\mathcal{E}) \ll \tau(C^*)$), coordination is robust to agent failure. The framework thus identifies the exact regime boundary (coordination strength $K > K_c = 2\sigma_\omega/\pi$) separating FLP-style impossibility from tractable consensus, consistent with the synchronization analysis of [32].

16.6 Mathematical Foundations

Banach’s fixed-point theorem [5] is the core existence tool, replacing Kakutani’s theorem. Borel [7] and Cantelli [8] proved the lemma characterizing infinite sums of probabilities. Halmos [14] and Kolmogorov [17] provide measure-theoretic foundations. Rudin [31] and Kelley [16] provide real-analysis and topological foundations. Munkres [26] provides the topology. Mac Lane [23] provides the categorical framework for the goal-quotient functor. Gödel’s incompleteness [12] establishes absolute limits on formal systems; the floor theorem can be seen as an agent-theoretic analogue: no bounded receiver has $\beta = 0$, just as no formal system is both complete and consistent. March’s exploration/exploitation framework [24] connects to the agent flow: exploitation corresponds to the

contractive phase (decreasing S), exploration to the expansive phase (searching for lower- S cells). Savage’s foundations of statistics [33] underlie the Bayesian interpretation of receiver priors.

17 Conclusion

This paper has developed a complete mathematical theory of market coordination among bounded heterogeneous agents. The ten principal contributions are:

- (i) **Belief Incompatibility Theorem** (theorem 3.2): representation-disjoint agents have structurally incompatible beliefs at every state — a universal property, not an empirical regularity.
- (ii) **Common-Cell Convergence Theorem** (theorems 4.1 and 4.2): despite full belief incompatibility, every agent in an ensemble can simultaneously attain the same action-cell, because cells live in outcome space, not representation space.
- (iii) **Multi-Agent Catalytic Composition Law** (theorem 5.2): the composite floor of n independent agents is $\prod_i S_b(A_i)/\Sigma^{n-1}$ — a product law on dimensionless floors.
- (iv) **Purpose Existence Theorem** (theorem 8.1): every independent ensemble with sufficient cell tolerance has a purpose equilibrium, proved by Banach contraction without convexity.
- (v) **Purpose Stability** (theorems 8.5 and 8.6): the equilibrium cell is preserved under ε -perturbations of the ensemble, with an explicit quantitative stability bound.
- (vi) **ω -Limit Theorem** (theorem 9.2): continuous agent flows starting in the reachability set converge to the purpose cell — markets clear dynamically.
- (vii) **Motivation Heterogeneity Theorem** (theorem 10.2): the composite floor is goal-content-invariant — buyers and sellers, bulls and bears, compose to the same information-aggregation floor.
- (viii) **$\boxtimes$ Algebra** (theorems 11.2 and 11.5): $([0, \Sigma], \boxtimes, \Sigma)$ is a commutative monoid; catalytic composition is the monoid operation;

the goal-quotient functor is a monoid homomorphism.

- (ix) **Market Information Efficiency** (theorems 13.2 to 13.4): the EMH holds iff $\sum_i \log(1/q_i) = \infty$; heterogeneity drives efficiency; perfect efficiency requires infinite heterogeneous participation.
- (x) **Coordination without Common Knowledge** (theorems 4.4 and 4.5): the Aumann common-knowledge hierarchy is replaced by cell exteriority — each agent reaches the equilibrium cell through its own decoder-projection chain independently.

This paper, together with [32], completes the mathematical foundation for computing economics with bounded agents: the single-agent floor theory in the companion paper, and the multi-agent equilibrium theory here. The framework requires no convexity, no common knowledge, no complete markets, no continuous preferences, and no rational expectations — only the primitive of a bounded receiver agent triple and the algebra of floor composition.

Forty-five numerical experiments verify all claims to machine precision (maximum relative error $\leq 10^{-14}$, table 1).

The companion paper on heterogeneity-driven market efficiency (Paper 3 of this series) will extend the present results to infinite ensembles, study the spectral theory of the purpose functional $\Phi_{\mathcal{E}}$ as an operator on the space of cells, and characterize the universality class of markets that satisfy the Borel–Cantelli efficiency condition.

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